MIL-L-20599
17 January 1952

MILITARY SPECIFICATION

LOADING WITH INERT FILLERS, MARKING AND PACKING

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments, and supersedes the following specification:

Army PXS-1238
31 August 1949

This specification consists of this cover sheet and Specification PXS-1238, dated 31 August 1949, attached hereto, without modification.

When a request for this specification is received by a supplying activity it will be necessary to attach this cover sheet to the pertinent specification before issue.

Copies of specifications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.

Custodian:
Army-O

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PXS-1238
31 August 1949

PICATINNY ARSENAL TENTATIVE SPECIFICATION

LOADING WITH INERT FILLERS, MARKING AND PACKING

1. SCOPE AND CLASSIFICATION

1.1 This specification covers the loading of standard items of ammunition with inert fillers to simulate various high-explosive fillers differing in specific gravities.

1.2 Compositions.—Inert fillers covered by this specification shall be of the following compositions:

Filler A - mixture of gypsum cement and water

Filler B - mixture of crude polychlornaphthalene and diatomaceous earth.

Filler C - mixture of crude polychlornaphthalene/diatomaceous earth and hydrocarbon wax.

Filler D - mixture of crude polychlornaphthalene/diatomaceous earth and ground iron oxide (hematite).

2. APPLICABLE SPECIFICATIONS AND DRAWINGS

2.1 Specifications.—The following specifications, of the issue in effect on the date of invitation for bids and all drawings and specifications listed or referred to therein, form a part of this specification:

U. S. ARMY SPECIFICATIONS

50-11-149 Polychlornaphthalene, Crude
52-5-27 Gypsum Cement

PICATINNY ARSENAL TENTATIVE SPECIFICATIONS

PXS-1252 Diatomaceous Earth
PXS-1263 Hydrocarbon Wax (For Use in Inert Fillers)

(Copies of specifications and drawings should be obtained from the procuring agency or as directed by that agency. Both the title and identifying number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Material.—

3.1.1 The materials used in the mixing of the various fillers shall conform to the applicable specifications listed in paragraph 2.1. In addition the following material shall be of the class specified:

Hydrocarbon Wax - Picatinny Arsenal Tentative Specification PXS-1263, Class 1.

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3.1.2 The iron oxide used in Filler D shall be a ground iron oxide ore, not less than 95 percent of which shall pass through a No. 325 (44 micron) U. S. Standard Sieve. It shall contain not less than 50 percent of ferric oxide (Fe_2O_3).

3.2 Closure of item loaded.--Items loaded with Filler A shall be closed and sealed to prevent evaporation of moisture from the charge in long term storage.

3.3 The loading shall meet all requirements prescribed on drawings and in applicable specifications.

3.4 Composition of Fillers.--

3.4.1 Filler A.--Mixture of gypsum cement and water, with proportions of each ingredient adjusted to produce a filler of the weight prescribed on the applicable drawing. (See 6.1)

3.4.2 Filler B.--Mixture of crude polychlornaphthalene and a minimum of 10 percent diatomaceous earth, with proportions of each ingredient adjusted to produce a filler of the weight prescribed on the applicable drawing. (See 6.1)

3.4.3 Filler C.--Mixture of crude polychlornaphthalene and diatomaceous earth, containing not more than 9 parts of the former for each part of the latter, with hydrocarbon wax added in the proportion (0 to 40 percent as required) necessary to produce a filler of the weight prescribed on the applicable drawing. (See 6.1)

3.4.4 Filler D.--Mixture of crude polychlornaphthalene and diatomaceous earth, containing not more than 9 parts of the former for each part of the latter, with ground iron oxide (hematite) added in the proportion (0 to 25 percent as required) necessary to produce a filler of the weight prescribed on the applicable drawing. (See 6.1)

3.5 Workmanship.--

3.5.1 The item of ammunition to be loaded shall be free from grease and other foreign material. Bare and slightly rusty areas shall be permitted on the cavity walls provided there is no scale. The cleaning method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents used.

3.5.2 After loading and before painting, the exterior of the item of ammunition shall be clean and free from loading compostion, rust, grease, and dirt.

4. SAMPLING, INSPECTION AND TEST PROCEDURES

4.1 Lot.--A lot shall consist of twenty thousand items of ammunition, maximum. A smaller quantity may constitute a lot at the discretion of the inspector. Two thousand plus or minus 500 shall constitute the first lot manufactured on any contract except when a new contract immediately follows the manufacture of satisfactory material of the same character.

4.2 Inspection.--

4.2.1 Inspection shall be as specified in United States Army General Specification for Ammunition Except Small Arms Ammunition, No. 50-0-1.

4.2.2 Rotating Band.--The rotating band of each item of ammunition (where applicable) shall be inspected to insure that it has not been damaged by being nicked or burred.

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4.2.3 Gaging.--After the item of ammunition has been loaded, painted, and assembled, the contractor shall gage each item using a gage pertaining to the item under contract. If the item fails to enter the gage freely, it shall be rejected subject to correction and resubmission for test.

4.3 Tests.--There are no tests applicable to this specification.

5. PREPARATION FOR DELIVERY

5.1 Packing.--The packing shall be in accordance with the drawings and specifications for the item being loaded.

5.2 Labeling and marking.--Labeling and marking shall be as indicated on the applicable drawings and specifications for the item being loaded.

5.3 Data cards.--Data cards, 5 x 8 inches, made of commercial manila tag board, approximately 200 pounds ream weight, shall be prepared for each lot of ammunition in accordance with the form specified below. Quantities (approximately 50 cards per lot) and distribution of data cards shall be in accordance with instructions issued by the Office, Chief of Ordnance.

AMMUNITION DATA CARD

Kind:_____ Ammunition Lot No._____
Quantity in Lot_____
Drg. No.*_____ Spec. No.*_____
Loaded by_____ Date_____
Contract No._____ Order No._____
Packed_____ _____in a box.
Remarks:_____

COMPONENTS

Kind	Mfr	Lot No.	Year Mfd	Date Filled	Loading Method	Drg. No.*
Projectile**						
Projectile Filler**						
Fuze**						
Fuzewell Cup**						
Rocket**						
Bomb**						

Certified to by_____Inspector.
Print here name of inspector, district, arsenal, or loading plant and subdistrict where applicable.

* Include Revision and Amendments
** Use only if applicable to item loaded

6. NOTES

6.1 The fillers covered by this specification may be used for simulation of various explosive fillers which differ widely in specific gravity. To assist in determining the percentage of ingredients of the respective fillers necessary to meet the specific gravity requirement specified on the

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applicable drawing, reference may be made to Figures 1 and 2. These figures indicate the specific gravity produced by various combinations of filler compositions, and also show the specific gravity of the more common standard high-explosive fillers. Experience has indicated that to insure ease of loading Fillers B, C, and D, the proportion of diatomaceous earth to polychloronaphthalene should not exceed 15 percent.

6.2 The addition of not more than .01 gram of Sodium Citrate for each pound of gypsum cement in Filler A is permissible to increase the time of pourability by one minute.

6.3 Notice.--When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever, and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications for other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

Custodian:

Picatinny Arsenal, Dover, N.J.

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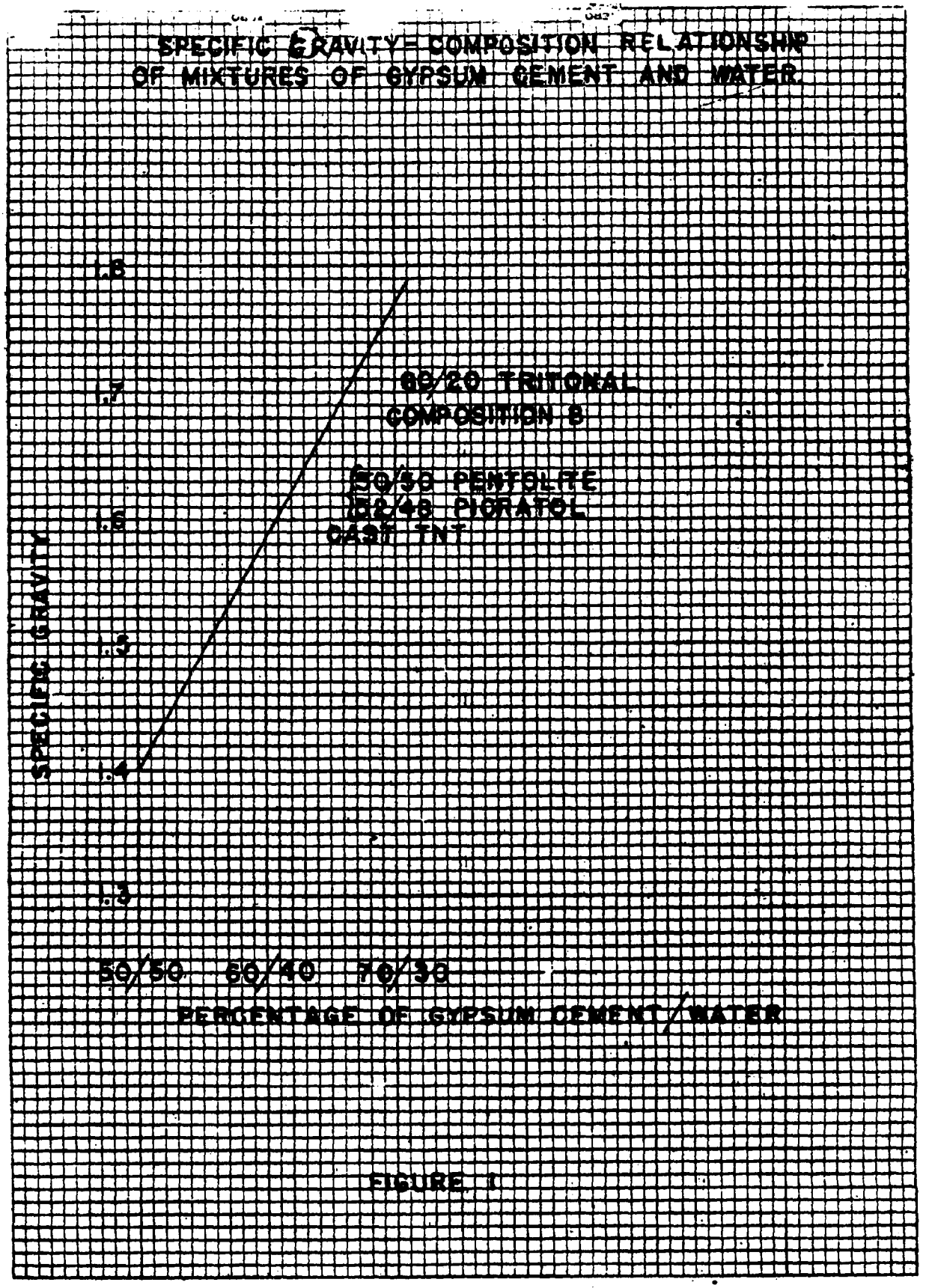


FIGURE 1

ADVISORY DATA

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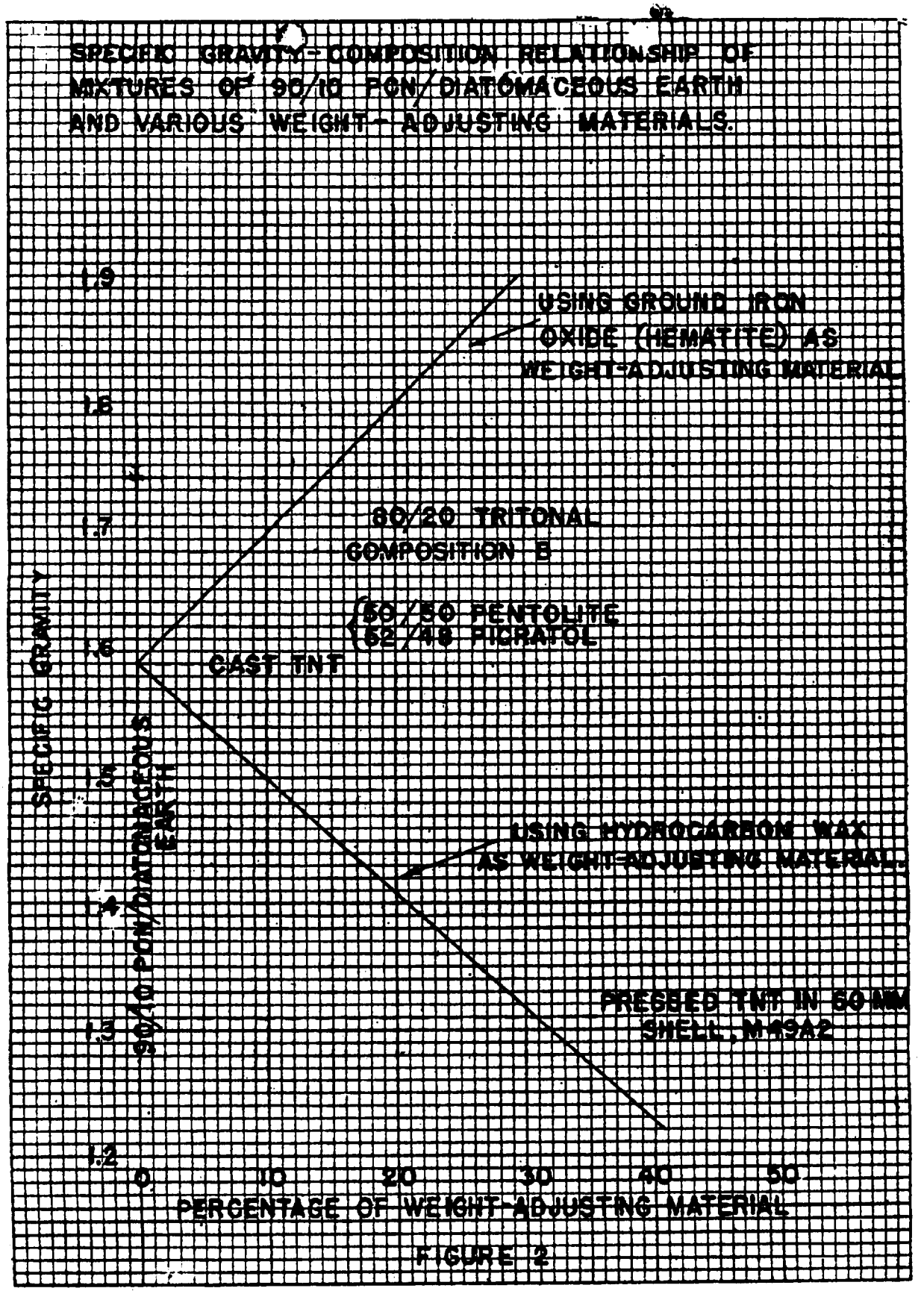


FIGURE 2

ADVISORY DATA

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD, FORM 1 APR 65 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

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